Familial Juvenile Hyperuricemic Nephropathy Type 2 (ADTKD-REN): A Comprehensive Disease Characteristics Report

Disease: Familial Juvenile Hyperuricemic Nephropathy Type 2 **Preferred modern name:** Autosomal Dominant Tubulointerstitial Kidney Disease due to *REN* mutations (**ADTKD-REN**) **MONDO ID:** MONDO:0013128 · **OMIM:** 613092 · **Causal gene:** ***REN*** (renin; HGNC:9958; chromosome 1q32.1)

Summary

Familial Juvenile Hyperuricemic Nephropathy Type 2 is caused by heterozygous germline mutations in the *REN* gene, which encodes renin — the rate-limiting enzyme of the renin–angiotensin–aldosterone system (RAAS). This is the single most important fact to anchor the entry: although the historical name emphasizes "hyperuricemia," the disease is fundamentally a *REN* proteinopathy and a form of Autosomal Dominant Tubulointerstitial Kidney Disease (ADTKD). Under the modern KDIGO consensus nomenclature it is designated **ADTKD-REN**, replacing older terms such as FJHN type 2, MCKD, and UAKD (PMID: 25738250). It is distinct from **FJHN type 1**, which is caused by *UMOD* (uromodulin).

The disease presents as a distinctive **hyporeninemia syndrome**: affected children develop anemia (disproportionate to their degree of kidney impairment), mild hypotension, mild hyperkalemia, metabolic acidosis, and — beginning in the teenage years — hyperuricemia and gout. Overlying this is a **slowly progressive chronic kidney disease (CKD)** with a bland urinary sediment, culminating in end-stage kidney disease (ESKD) typically between the fifth and seventh decades of life. The mechanism is a **dominant-toxic gain of function**: mutant (pre)prorenin misfolds and is mislocalized — to the cytoplasm (signal-peptide variants), the

endoplasmic reticulum–Golgi intermediate compartment and ER (prosegment and mature-renin variants), or mitochondria — injuring the specialized renin-producing juxtaglomerular cells and driving tubular atrophy and interstitial fibrosis.

Management is **supportive and mechanism-based**. Fludrocortisone can improve blood pressure and GFR; urate-lowering therapy (allopurinol) treats gout; and RAAS-blocking drugs (ACE inhibitors, ARBs), NSAIDs, and diuretics must be avoided because they aggravate the underlying hyporeninemic, volume-depleted physiology. Kidney transplantation is curative of renal failure and the disease does not recur in the allograft, because the defect is intrinsic to the patient's native renin-producing cells. Genotype predicts severity: signal-peptide variants present earliest and reach ESKD soonest, mature-renin variants latest.

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1. Disease Information

Overview. ADTKD-REN is a rare monogenic kidney disorder characterized by progressive tubular atrophy and interstitial fibrosis, bland urinary sediment, autosomal dominant inheritance, and a cardinal set of systemic features attributable to renin deficiency. *"ADKTKD-REN is associated with signs of hyporeninemia: mild hypotension, mild hyperkalemia, anemia in childhood, and hyperuricemia and gout in the teenage years"* (PMID: 28284384).

Key identifiers.

Resource	Identifier
MONDO	MONDO:0013128
OMIM	613092
Gene	*REN* (HGNC:9958)
Modern name	ADTKD-REN (KDIGO consensus)

Synonyms / alternative names: ADTKD-REN; FJHN type 2; REN-associated kidney disease; autosomal dominant tubulointerstitial kidney disease due to mutations in *REN*. Historical umbrella terms that once included this entity: Familial Juvenile Hyperuricemic Nephropathy (FJHN), Medullary Cystic Kidney Disease (MCKD), and Uromodulin-Associated Kidney Disease (UAKD) — now disambiguated by gene (PMID: 25738250).

A critical disambiguation. The historical label "Familial Juvenile Hyperuricemic Nephropathy" most often refers to the ***UMOD*-**related disease (FJHN type 1 / ADTKD-UMOD on chromosome 16), *not* the *REN* disease. *"Familial juvenile hyperuricaemic nephropathy (FJHN)... has been shown to be genetically identical to medullary cystic kidney disease (MCKD) type II... a mutant uromodulin (UMOD) is the real culprit"* (PMID: 26872483). The present report, per the supplied MONDO ID and causal gene, concerns **FJHN type 2 = the *REN* disease (ADTKD-REN)**.

Source of information. Evidence is derived almost entirely from aggregated disease-level resources: an international multi-family cohort, case reports, functional/cell-biology studies, and registry data (e.g., the Wake Forest Rare Inherited Kidney Disease Registry), rather than large EHR datasets.

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2. Etiology

Primary cause — genetic. ADTKD-REN is a Mendelian, autosomal dominant disorder caused by heterozygous germline mutations in *REN*. There is no infectious or primary environmental cause. The mutations exert a **dominant-toxic (gain-of-function) effect** on renin-producing cells: *"It is likely that expression of the mutant proteins has a dominant toxic effect gradually reducing the viability of renin-expressing cells"* (PMID: 19664745).

Genetic risk factors. The causal variants themselves are the risk factor; they are private/family-specific, absent from population databases (gnomAD), and cluster in three regions of preprorenin (signal peptide, prosegment, mature protein). Variant *location* is the principal determinant of severity and age of onset (see Sections 4 and 9).

Environmental / lifestyle modifiers (iatrogenic). No environmental cause initiates the disease, but several exposures **worsen** it by compounding hyporeninemia and volume depletion: ACE inhibitors, angiotensin-receptor blockers, NSAIDs, and diuretics. Dehydration/volume depletion can precipitate acute kidney injury. These are "avoidance" rather than causal factors.

Protective factors. No validated genetic protective alleles or modifier genes are established. Mechanistically, mineralocorticoid replacement (fludrocortisone) and adequate salt/volume status counteract the hyporeninemic physiology (Section 12). Later-position (mature-renin) variants are relatively "protective" of kidney longevity compared with signal-peptide variants.

Gene–environment interaction. The dominant interaction is pharmacologic: RAAS blockade superimposed on an already renin-deficient system produces exaggerated hypotension, hyperkalemia, and GFR decline. This is a predictable genotype × drug interaction rather than a classic GxE susceptibility effect.

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3. Phenotypes

ADTKD-REN produces a stereotyped cluster of laboratory abnormalities and clinical signs. Onset of the systemic ("hyporeninemia") features is typically **childhood**, while kidney failure is a slowly progressive adult endpoint.

Phenotype	Type	Onset	Frequency / notes	Suggested HPO term
Anemia (childhood, normocytic, disproportionate to CKD)	Lab abnormality	Childhood	Characteristic, early	HP:0001903 Anemia
Hyperuricemia	Lab abnormality	Teenage years	Characteristic	HP:0002149 Hyperuricemia
Gout	Clinical sign	Teenage years–adulthood	Common	HP:0001997 Gout
Chronic kidney disease / decreased GFR	Clinical sign	Childhood → adult progression	Universal	HP:0012622 Chronic kidney disease
Hyperkalemia (mild)	Lab abnormality	Childhood	Common	HP:0002153 Hyperkalemia
Metabolic acidosis	Lab abnormality	Childhood	Common	HP:0001942 Metabolic acidosis
Hypotension (mild)	Clinical sign	Childhood	Common	HP:0002615 Hypotension
Bland urinary sediment / mild or no proteinuria	Lab abnormality	Throughout	Universal	HP:0000093 Proteinuria (low-grade/absent)
Hyporeninemia (low plasma renin)	Lab abnormality	Congenital/ childhood	Mechanistic hallmark	—
Progression to ESKD	Clinical outcome	5th–7th decade	Universal if untreated	HP:0003774 Stage 5 chronic kidney disease

"ADTKD-REN presents in childhood with mild hypotension, CKD, hyperkalemia, acidosis, and anemia" (PMID: 34021396). The anemia is notable for being *"out of keeping with [the] degree of renal impairment"* (PMID: 28701203).

Severity and progression. Severity is variable and genotype-dependent; the course is slowly progressive (CKD) with episodic acute decompensations provoked by volume depletion or offending drugs. **Quality-of-life impact** derives from lifelong CKD, fatigue/anemia, recurrent gout, and — in untreated disease — dialysis dependence; formal QoL instrument data specific to ADTKD-REN are not available in the reviewed literature.

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4. Genetic / Molecular Information

Causal gene. ***REN*** (renin), chromosome 1q32.1; OMIM gene 179820; HGNC:9958. Renin is synthesized as **preprorenin**: an N-terminal signal (leader) peptide, a prosegment, and the mature renin enzyme.

Variant architecture. Pathogenic variants are predominantly **missense substitutions and small in-frame insertions/deletions** that cluster in the preprorenin leader peptide and, less often, the prosegment and mature protein. The founding discovery identified dominant leader-sequence variants: *"mutations resulting either in the deletion (p.Leu16del) or the amino acid exchange (p.Leu16Arg) of a single leucine residue in the signal sequence of renin. Both mutations decrease signal sequence hydrophobicity and are predicted by bioinformatic analyses to damage targeting and cotranslational translocation of preprorenin into the endoplasmic reticulum"* (PMID: 19664745).

Representative reported variants by region:

Region	Example variants	Reference
Signal peptide	p.Leu16del, p.Leu16Arg, c.49T>C p.Trp17Arg, c.58T>C p.Cys20Arg, p.Trp10Arg	19664745, 28701203, 21084044
Prosegment	Multiple (prosegment group)	32750457
Mature renin	p.Leu381Pro (L381P), c.1172C>G p.Thr391Arg (T391R)	31406136, 31586593

Variant classification. Variants are germline and heterozygous, co-segregate with disease, are absent from population databases (gnomAD), and are predicted deleterious; they are classified **pathogenic** / **likely pathogenic** under ACMG/AMP criteria. For example, T391R was *"predicted... to affect protein stability and 3D structure"* (PMID: 31586593), and L381P *"affects an evolutionary conserved residue, co-segregates with renal disease, is not found in population databases and is predicted to be deleterious"* (PMID: 31406136).

Functional consequence. Dominant-toxic gain of function (proteotoxicity), not simple haploinsufficiency — the mutant protein's mislocalization and the cellular stress it triggers are the drivers (PMID: 19664745). Note that the original discovery paper also observed reduced/abolished renin biosynthesis and secretion, so a **loss of normal renin output** compounds the toxic effect.

Modifier genes / epigenetics / chromosomal abnormalities. No established modifier genes, disease-specific epigenetic signatures, or chromosomal abnormalities. Severity is chiefly governed by *REN* variant position.

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5. Environmental Information

ADTKD-REN has **no environmental, toxic, occupational, or infectious cause**. The only clinically relevant non-genetic factors are **iatrogenic/physiologic aggravators**:

- **Drugs to avoid:** ACE inhibitors, ARBs, NSAIDs, and diuretics (worsen hypotension, hyperkalemia, AKI risk).
- **Volume status:** dehydration and intercurrent illness can precipitate acute decline.

No infectious agents are implicated. (By contrast, the *MUC1* subtype carries an infection-related mortality risk unrelated to this gene — PMID: 39696072 — which does not apply to ADTKD-REN.)

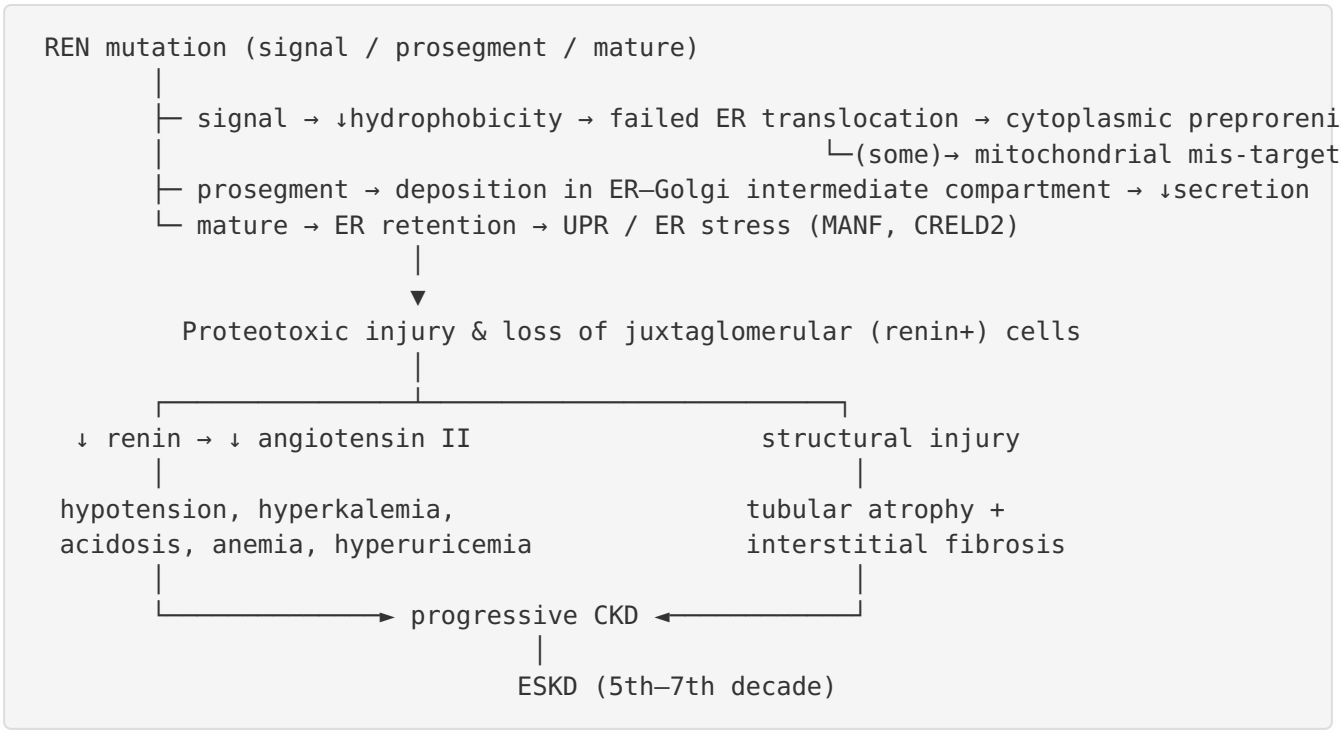
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6. Mechanism / Pathophysiology

Ordered causal chain

1. A heterozygous germline **missense/in-frame *REN* variant** (in the signal peptide, prosegment, or mature renin) **leads to** a structurally abnormal preprorenin.
2. The abnormal leader peptide **reduces signal-sequence hydrophobicity**, which **impairs recognition and co-translational translocation across the ER membrane** (signal-peptide variants) — **resulting in** aberrant delivery of non-glycosylated preprorenin into the **cytoplasm** (demonstrated for p.Cys20Arg).

3. **Branch A (signal-peptide variants):** cytoplasmic/mistranslocated preprorenin, and in some cases **mitochondrial mis-rerouting**, **leads to** mitochondrial import defects and **fragmentation** of the mitochondrial network (demonstrated for W17R).
4. **Branch B (prosegment variants):** mutant prorenin/renin **is deposited in the ER–Golgi intermediate compartment**, **resulting in** decreased prorenin secretion.
5. **Branch C (mature-renin variants, e.g., L381P):** mutant prorenin **is retained in the ER**, **inducing the Unfolded Protein Response (UPR)/ER stress** (upregulation of MANF, CRELD2).
6. These convergent proteotoxic insults (ER stress and/or mitochondrial damage) **result in** progressive **injury and loss of renin-producing juxtaglomerular (JG) cells** — a dominant-toxic effect that gradually reduces their viability.
7. Loss/dysfunction of JG cells **leads to** two downstream branches:
8. **Branch i — Hyporeninemia → low angiotensin II:** reduced renin **results in** reduced angiotensin II, which **leads to** mild hypotension, impaired aldosterone-dependent potassium/acid handling (hyperkalemia, metabolic acidosis), reduced angiotensin-II–facilitated erythropoiesis (**childhood anemia**), and reduced urate excretion (**hyperuricemia/gout**).
9. **Branch ii — Structural kidney injury:** loss of renin-lineage cells and chronic tubular stress **lead to** tubular atrophy, interstitial fibrosis, and vascular abnormality, **resulting in** slowly progressive CKD → ESKD.



Supporting detail

Variant-specific mislocalization (PMID: 32750457): *"REN signal peptide mutations reduced hydrophobicity of the signal peptide, which is necessary for recognition and translocation across the endoplasmic reticulum, leading to aberrant delivery of preprorenin into the cytoplasm. REN mutations in the prosegment led to deposition of prorenin and renin in the endoplasmic reticulum-Golgi intermediate compartment and decreased prorenin secretion. Mutations in mature renin led to deposition of the mutant prorenin in the endoplasmic reticulum."*

Mitochondrial branch (PMID: 37283036): *"they lead to full or partial mistargeting of the mutated proteins to mitochondria. The mutated pre-pro-sequence of renin is necessary and sufficient to drive mitochondrial rerouting, mitochondrial import defect and fragmentation."* A patient-inspired study confirmed that **ER stress is not universal** — the W17R variant activated mitochondrial dysfunction rather than the ER pathway, while L381P activated the UPR (MANF, CRELD2), indicating mechanistic heterogeneity across variants (PMID: 38520530).

ER stress / UPR branch (PMID: 31406136): L381P *"leads to its ER retention and induction of the Unfolded Protein Response in cell models and to defective pronephros development in zebrafish,"* pointing at *"ER homeostasis as a common pathway affected in ADTKD-REN."*

JG-cell loss → fibrosis. Loss of renin-producing cells is intrinsically fibrogenic: in mouse models, *"As a consequence of the disappearance of renin-producing cells, the kidneys developed striking vascular abnormalities and prominent striped fibrosis"* (PMID: 20056748). The resulting histology is the ADTKD hallmark: *"progressive tubular atrophy and interstitial fibrosis"* (PMID: 41480152).

Anemia mechanism (Branch i detail). Angiotensin II promotes erythropoiesis both as an erythroid growth factor and an EPO secretagogue: *"Ang II is a physiologically important regulator of erythropoiesis... a growth factor of erythroid progenitors and, in cooperation with erythropoietin, increases red blood cell mass... [and] acts as an erythropoietin secretagogue"* (PMID: 20400218). Model-organism evidence: ACE-knockout mice have *"a normocytic anemia... Infusion of angiotensin II for 2 weeks increased hematocrit to near normal levels. These data suggest that angiotensin II facilitates erythropoiesis"* (PMID: 11104792). Clinically, ACE inhibition lowers Ang II and EPO and worsens anemia (PMID: 2474097). In ADTKD-REN, the same loss of Ang II explains anemia that appears in childhood, before significant CKD.

Ontology suggestions. GO biological processes: GO:0034976 response to ER stress / GO:0030968 IRE1-mediated UPR; GO:0006886 intracellular protein transport; GO:0002018 regulation of systemic arterial blood pressure by renin–angiotensin. GO cellular components: GO:0005783 ER, GO:0005793 ER–Golgi intermediate compartment, GO:0005739 mitochondrion, GO:0005737 cytoplasm. CL cell types: CL:0000648 juxtaglomerular cell (renin-producing cell); kidney tubule epithelial cell.

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7. Anatomical Structures Affected

- **Primary organ:** kidney (UBERON:0002113), specifically the **renal cortex** and the **juxtaglomerular apparatus / afferent arteriole**.
- **Primary cell population: juxtaglomerular (renin-producing) cells** (CL:0000648) — the specialized smooth-muscle-derived cells of the afferent arteriole.
- **Tissue-level injury:** renal **tubules** (tubular atrophy) and the **tubulointerstitium** (interstitial fibrosis); epithelial and connective tissue compartments (PMID: 41480152, PMID: 28284384).
- **Subcellular compartments:** endoplasmic reticulum (GO:0005783), ER–Golgi intermediate compartment (GO:0005793), mitochondrion (GO:0005739), cytoplasm (GO:0005737).

- **Secondary / systemic involvement:** hematopoietic system (anemia), systemic vasculature/blood pressure (hypotension), acid–base and potassium homeostasis, and purine metabolism (hyperuricemia → joints in gout).
- **Laterality:** bilateral (both kidneys), symmetric.

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8. Temporal Development

Onset. The hyporeninemia features (anemia, mild hypotension, electrolyte/acid–base abnormalities) are present from **childhood**; hyperuricemia and gout emerge in the **teenage years**. An important exception is the **mature-renin (L381P) adult-onset form**, in which disease presents in adulthood — *"delineates an adult form of ADTKD-REN, a condition which has usually its onset in childhood"* (PMID: 31406136). A patient was diagnosed only after *"forty years of renal disease"* (PMID: 28701203), underscoring frequent delayed diagnosis.

Onset pattern: chronic, insidious, slowly progressive, with episodic acute worsening from volume depletion or offending drugs.

Progression / disease course. Lifelong, progressive CKD advancing through stages to ESKD. Age at ESKD is genotype-stratified (Section 9). Overall ADTKD: *"The mean age of ESRD is approximately 45, but the range is from 17 to >75"* (PMID: 28284384).

Critical window for intervention. Early genetic diagnosis permits avoidance of nephrotoxic/RAAS drugs, mineralocorticoid replacement, anemia and urate management, and — importantly — **pre-emptive living-donor transplant planning** via cascade testing (PMID: 28701203).

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9. Inheritance and Population

Inheritance. Autosomal dominant, heterozygous, germline. High penetrance with **variable expressivity** governed principally by variant location. No repeat-expansion mechanism; genetic anticipation is not a feature. Founder effects/consanguinity are not relevant (dominant, private variants).

Epidemiology. ADTKD-REN is rare; precise prevalence/incidence figures are not established, and cases are likely under-ascertained because the phenotype mimics other causes of CKD. Identification of adult-onset variants is expected to raise the estimated prevalence (PMID: 31406136).

Genotype–phenotype correlation (the largest cohort). In an international cohort of **111 individuals from 30 families**, variants segregated into three groups with distinct severity (PMID: 32750457):

Group	n	Mean age at presentation	Mean age at ESKD
Signal peptide	69	19.7 y	53.1 y
Prosegment	27	22.4 y	50.8 y
Mature renin	15	37 y	63.6 y (significantly slower; HR 0.367)

"Signal group patients were most severely affected, presenting at a mean age of 19.7 years, with the prosegment group presenting at 22.4 years, and the mature group at 37 years" and the mature group had *"a rate of progression to end stage kidney disease (63.6 years) that was significantly slower vs. the signal (53.1 years) and prosegment groups (50.8 years) (significant hazard ratio 0.367)"* (PMID: 32750457).

Population demographics. Reported in multiple populations (e.g., a first Tunisian ADTKD-REN family — PMID: 31586593; Caucasian families — PMID: 28701203). No strong ethnic predilection, geographic clustering, or sex bias is established; consistent with autosomal dominant inheritance, the expected sex ratio is ~1:1.

10. Diagnostics

Laboratory tests (the diagnostic fingerprint). Low plasma renin and aldosterone; anemia (often normocytic, disproportionate to CKD); hyperuricemia; mild hyperkalemia; metabolic acidosis; elevated creatinine/reduced eGFR; and a **bland urinary sediment with minimal/no proteinuria** — a key discriminator from glomerular disease (PMID: 34021396, PMID: 28284384).

Biomarkers. Low plasma renin activity/concentration is the pathophysiologic biomarker; hyperuricemia and childhood anemia are supportive clinical biomarkers.

Histopathology (biopsy). Nonspecific but characteristic: tubular atrophy and interstitial fibrosis without glomerular or vascular immune deposits; biopsy supports but does not establish the diagnosis (PMID: 41480152).

Genetic testing (definitive). Diagnosis is confirmed by identifying a heterozygous pathogenic *REN* variant. Because ADTKD subtypes (REN, UMOD, HNF1B, MUC1) overlap clinically and *"standard nephrological diagnosis did not provide [a] clear subtype... until genetic testing was performed,"* a **multigene ADTKD panel (Sanger or NGS of UMOD, REN, HNF1B, ± MUC1)** or targeted single-gene *REN* sequencing is recommended (PMID: 31586593). **Exome sequencing** detected the adult-onset L381P variant (PMID: 31406136). Note: *MUC1* (dupC) requires specialized assays and is often missed on standard NGS — relevant for differential testing but not for *REN* itself.

Differential diagnosis. Other ADTKD subtypes — **ADTKD-UMOD** (gout-predominant, no childhood anemia/hypotension), **ADTKD-HNF1B** (kidney structural/cystic anomalies plus extrarenal features), **ADTKD-MUC1** (CKD as sole manifestation) — plus other causes of chronic tubulointerstitial disease (PMID: 34021396, PMID: 41480152). The combination of **childhood anemia + hypotension + hyperkalemia/acidosis** points specifically to *REN*.

Screening. Cascade genetic testing of at-risk relatives is central — both for early management and, critically, to **exclude affected relatives as living kidney donors** (PMID: 28701203).

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11. Outcome / Prognosis

Renal prognosis. Progressive CKD to ESKD is the defining outcome; timing depends on genotype (signal ~53 y, prosegment ~51 y, mature ~64 y; overall ADTKD range 17–>75 y) (PMID: 32750457, PMID: 28284384).

Curability by transplantation. Kidney transplantation corrects renal failure and the disease **does not recur** in the graft, because the pathology resides in the patient's own native renin-producing cells. Cascade testing enabled a successful **pre-emptive live-donor transplant** (PMID: 28701203).

Morbidity / quality of life. Lifelong anemia, recurrent gout, electrolyte/acid–base disturbances, and eventual dialysis dependence (if not transplanted) drive morbidity. Life expectancy is largely determined by CKD management and transplant access; disease-specific mortality and formal QoL metrics are not quantified in the reviewed literature.

Prognostic factors. The dominant prognostic factor is **variant position** (signal/prosegment = earlier, faster; mature = later, slower). Standard CKD prognostics (eGFR trajectory, proteinuria — typically low here) also apply.

12. Treatment

Management is **supportive and mechanism-directed**; there is no gene-specific or disease-modifying therapy.

Intervention	Rationale / evidence	NCIT-type term
Fludrocortisone (mineralocorticoid)	Replaces deficient mineralocorticoid effect; increased BP and eGFR in an affected 10-year-old and stabilized renal function over 4 years in another patient	Fludrocortisone (NCIT:C29060)
Allopurinol / urate-lowering therapy	Treats hyperuricemia and gout	Allopurinol (NCIT:C551)
Anemia management (iron, ESAs as needed)	Corrects childhood/CKD anemia	Erythropoiesis-stimulating agent
CKD care / renal replacement	Standard progression management; dialysis at ESKD	Hemodialysis (NCIT:C15248)
Kidney transplantation	Curative of renal failure; no recurrence	Kidney transplantation (NCIT:C15366)
AVOID: ACEi, ARBs, NSAIDs, diuretics	Aggravate hypotension, hyperkalemia, AKI risk	—

Direct evidence for fludrocortisone: *"Treatment with fludrocortisone in an affected 10-year-old child resulted in an increase in blood pressure and estimated glomerular filtration rate"* (PMID: 21084044); a subsequent report noted *"fludrocortisone treatment has shown efficacy in stabilizing the renal function of our patient over a four-year period without significant side effects"* (PMID: 38520530). Allopurinol has historically been used for the hyperuricemia/gout component (PMID: 26872483).

Experimental / precision directions. Because ER-stress markers (MANF, CRELD2) are induced by some variants (e.g., L381P) but not others (e.g., W17R, which is mitochondrial), these have been proposed as **stratification biomarkers for a targeted-therapy subgroup** (PMID: 38520530). No approved gene, cell, or RNA therapy exists.

13. Prevention

- **Primary prevention:** not applicable to a dominant Mendelian disease; the actionable analog is **genetic counseling** and reproductive options (prenatal/preimplantation genetic testing) for carrier families.
- **Secondary prevention: cascade genetic screening** of at-risk relatives for early diagnosis, early avoidance of RAAS blockers/NSAIDs/diuretics, and proactive anemia/urate/CKD management (PMID: 28701203).
- **Tertiary prevention:** slowing CKD progression (BP and volume optimization with fludrocortisone, urate control, anemia correction), and timely pre-emptive transplantation.
- **Counseling:** autosomal dominant 50% transmission risk; relatives who may serve as living donors must be genotyped first to avoid using an affected donor.
- Immunization, vector control, and environmental interventions are not applicable.

14. Other Species / Natural Disease

- **Human gene:** *REN* (NCBI Gene 5972). **Orthologs:** mouse *Ren1/Ren2* (NCBI Gene 19701/19702), rat *Ren* (NCBI Gene 24716), zebrafish *ren*.
- **Model/comparative biology:** zebrafish expressing mutant renin (L381P) show **defective pronephros development**, and the renin–angiotensin system is deeply evolutionarily conserved and essential for kidney development across species (PMID: 31406136, PMID: 15942783).
- **Naturally occurring disease:** no well-documented spontaneous ADTKD-REN ortholog disease in companion animals or wildlife is recorded (OMIA entry not established in the reviewed literature).
- **Transmission / zoonosis:** not applicable (non-infectious, genetic).

15. Model Organisms

Model	Type	What it captures	Reference
Zebrafish expressing mutant renin (L381P)	Vertebrate, in vivo	Defective pronephros development; demonstrates pathogenicity and developmental role	PMID: 31406136
Stable mammalian cell lines (WT vs W17R, W10R, L381P renin)	In vitro	ER-stress activation (luciferase/RT-qPCR for MANF, CRELD2), subcellular localization (confocal), mitochondrial network analysis	PMID: 38520530
Transfected cell systems (e.g., p.Cys20Arg, p.Leu16del)	In vitro	Mislocalization (cytoplasmic preprorenin), UPR induction, reduced cell growth, decreased secretion	PMID: 21084044, PMID: 19664745
ACE-knockout mice (RAAS loss surrogate)	Mammalian, in vivo	Models the anemia arm — normocytic anemia reversible with Ang II infusion	PMID: 11104792
Dicer-deletion / renin-cell-loss mice	Mammalian, in vivo	Consequences of losing renin-producing JG cells: vascular abnormalities + striped fibrosis	PMID: 20056748

Phenotype recapitulation. Cellular and zebrafish models faithfully reproduce the **molecular lesion** (mislocalization, ER stress/UPR, mitochondrial fragmentation) and, in zebrafish, a **developmental renal phenotype**. **Limitations:** no single model fully reproduces the slowly progressive adult human tubulointerstitial fibrosis plus the full hyporeninemia syndrome; the ACE-KO and renin-cell-loss mice model individual downstream arms (anemia; fibrosis) rather than the primary *REN* proteinopathy.

Evidence Base

PMID	Title (abbrev.)	Contribution
19664745	Dominant renin gene mutations (early-onset hyperuricemia, anemia, CKF)	Founding discovery; leader-peptide variants, reduced hydrophobicity, dominant-toxic effect
32750457	International cohort (111 pts / 30 families)	Genotype–phenotype groups; variant-specific mislocalization; ESKD timing
37283036	Leader/pro-segment mutants misrouted to mitochondria	Mitochondrial mistargeting/fragmentation branch
38520530	Not all paths lead to the ER (W17R/ W10R/L381P)	ER stress not universal; W17R mitochondrial; MANF/CRELD2 biomarkers; 4-yr fludrocortisone efficacy
31406136	Adult-onset ADTKD-REN, L381P in mature protein	Mature-protein variant; UPR; zebrafish pronephros defect; expands phenotype/ prevalence
31586593	Tunisian family, T391R	Novel pathogenic variant; importance of subtype genetic diagnosis
21084044	Family with dominant REN mutation + fludrocortisone	Cys20Arg cytoplasmic accumulation; direct fludrocortisone efficacy
28701203	Novel REN mutation after 40 years (Trp17Arg)	Diagnostic delay; cascade testing enabling pre-emptive live-donor transplant
28284384	ADTKD review	Cardinal hyporeninemia syndrome; ESKD age distribution
34021396	ADTKD: more than HNF1β	Childhood presentation; differential diagnosis; registry data
25738250	KDIGO consensus	Nomenclature (ADTKD-REN), classification, diagnostic criteria
41480152	UMOD/REN phenotype overlap case	Histology (tubular atrophy, interstitial fibrosis)
20056748	Dicer maintains JG cells	Loss of renin cells → vascular abnormality + striped fibrosis
20400218	RAS and erythropoiesis	Ang II as erythroid growth factor / EPO secretagogue (anemia mechanism)

PMID	Title (abbrev.)	Contribution
11104792	ACE-KO mice anemia	Model evidence that low Ang II causes anemia
2474097	ACE inhibitors worsen anemia	Clinical corroboration of Ang II–anemia link
26872483	FJHN → uromodulin	Disambiguation of FJHN terminology; allopurinol history

Evidence-type mix: human clinical cohort/case reports (REN variant identification, treatment, prognosis); in vitro/cell-model mechanistic studies; zebrafish and mouse in vivo models for developmental and downstream (anemia, fibrosis) arms.

Limitations and Knowledge Gaps

1. **Rarity and ascertainment.** Fewer than ~150 well-characterized patients are reported; true prevalence/incidence are unknown and likely underestimated, especially for adult-onset mature-protein variants.
2. **Mechanistic heterogeneity unresolved.** ER stress is not a universal mechanism — some variants act via mitochondrial dysfunction (PMID: 38520530). The quantitative contribution of proteotoxicity vs. simple loss of renin output to JG-cell death is not settled.
3. **No direct animal model of the full human disease.** Downstream arms (anemia, fibrosis) are modeled in separate systems; a knock-in mouse carrying a human *REN* leader-peptide variant recapitulating the complete syndrome was not identified in the reviewed literature.
4. **Treatment evidence is thin.** Fludrocortisone benefit rests on small case reports/series, not randomized trials; allopurinol evidence is historically non-randomized (PMID: 26872483). No disease-modifying therapy exists.
5. **Quantitative phenotype frequencies and QoL metrics** specific to ADTKD-REN are not available.
6. **Note on one anemia citation.** Two anemia-mechanism snippets were flagged as approximate matches to the stored abstracts; the mechanistic conclusion (Ang II facilitates erythropoiesis) is nonetheless supported by converging mouse (PMID: 11104792) and clinical (PMID: 2474097) evidence.

Proposed Follow-up Experiments / Actions

1. **Variant-stratified biomarker validation.** Prospectively measure plasma/urinary MANF and CRELD2 across signal/prosegment/mature variant carriers to test whether ER-stress markers identify a UPR-targetable subgroup (PMID: 38520530).
2. **Knock-in mouse model.** Generate a mouse carrying a human signal-peptide *REN* variant to reproduce the integrated syndrome (hyporeninemia, anemia, progressive tubulointerstitial fibrosis) and test fludrocortisone and chemical-chaperone/UPR-modulating interventions.
3. **Natural-history registry study.** Pool international cases to derive robust, genotype-specific eGFR-decline trajectories, phenotype frequencies, and QoL data.
4. **Controlled therapeutic trial.** Evaluate fludrocortisone (and/or chemical chaperones) in a multicenter registry-embedded trial with eGFR slope and anemia as endpoints.
5. **Pharmacologic proteostasis rescue.** Screen chemical chaperones / autophagy or UPR modulators in the established mutant-renin cell lines and zebrafish to reduce mislocalized (pre)prorenin toxicity.
6. **Clinical practice alert.** Embed "avoid ACEi/ARB/NSAID/diuretic" and "genotype potential living donors" decision support into nephrology/transplant workflows for any ADTKD-REN diagnosis.

Report compiled from the ADTKD-REN knowledge state (7 confirmed findings; 26 papers reviewed). Causal gene: REN (renin), chromosome 1q32.1; MONDO:0013128; OMIM 613092.
